Stereo Vision-Based Force Estimation and 3D Reconstruction Toward Autonomous Colorectal Anastomosis

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Motivation:

① Autonomous colorectal anastomosis requires reliable perception of tool–tissue interaction.

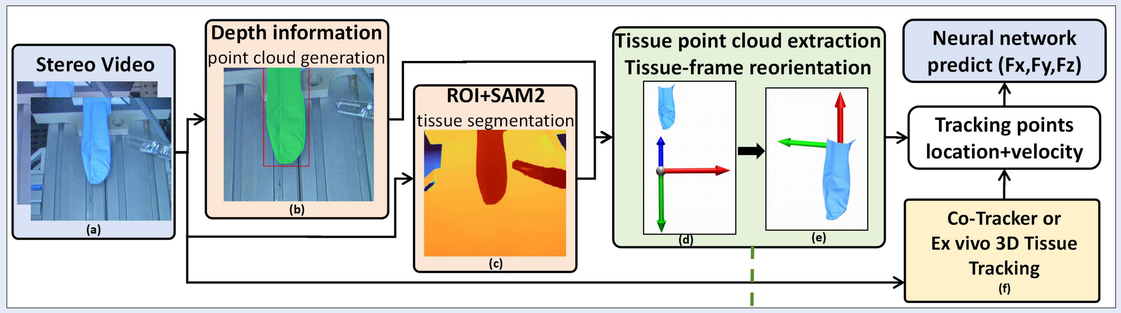
② Direct force sensing is often unavailable in surgical robotic systems.

③ We investigate 3D interaction-force estimation from stereo endoscopic video

④ Tissue tracking is unreliable during suturing due to reflections, weak texture, and occlusion.

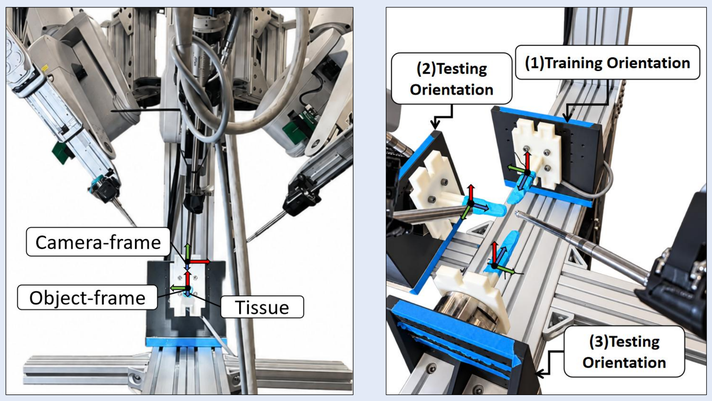
Method:

Vision-based 3D force estimation



Tissue-frame reorientation: **(1)** we combine stereo depth and tissue masks to reconstruct a 3D tissue point cloud. **(2)** we use PCA to define the initial longitudinal direction as the z-axis, and refine the colon centerline using the centers of elliptical cross-sections. **(3)** We use a fixed upward direction (opposite to gravity) in the ECM/world frame to define a consistent x-axis and determine the y-axis using the right-hand rule.

Experiments:

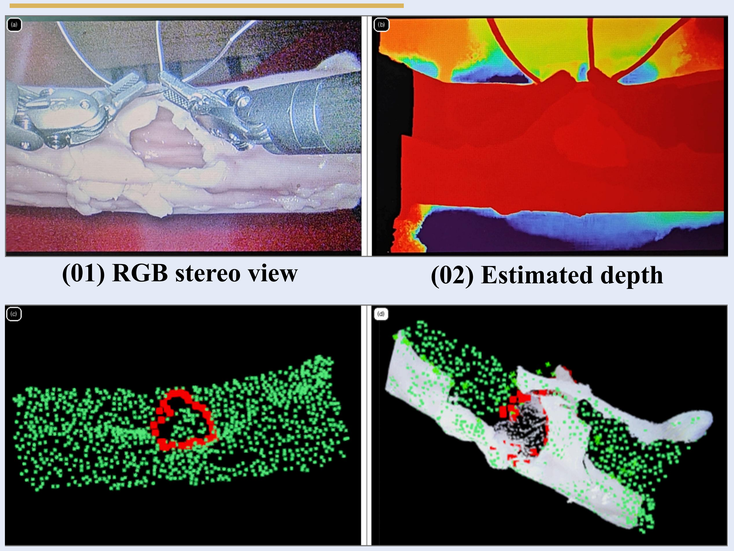


(a) (b)

Rubber-glove phantom: (a) Experimental setup. (b) Training, evaluation and testing orientations

Ex Vivo Colon Tracking Results:

3D Tissue Tracking for Ex Vivo Colon



(01) RGB stereo view **(02) Estimated depth**

(03) Wound identification (red) **(04) Partially occluded cloud**

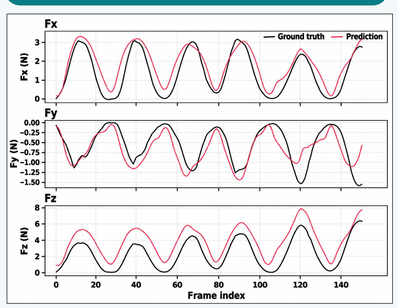
① Porcine colon suturing introduces specular reflections and instrument occlusion.

② We fit visible tracking points to visible tissue surface using a Chamfer-based objective. Local Linear Reconstruction (LLR) preserves neighboring structure and propagates deformation to occluded points.

Force Estimation & Ex Vivo Results:

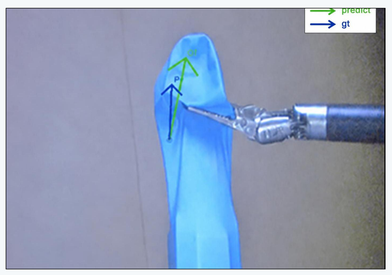
Held-out rubber-glove test

HELD-OUT GLOVE



Ground truth (black) vs. prediction (red)

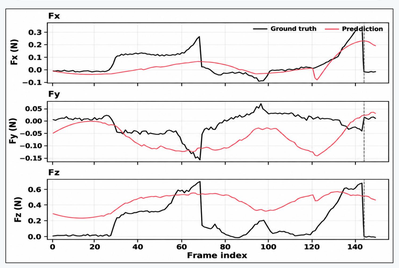
FORCE OVERLAY



Predicted force (green) overlaid with ground-truth force (blue).

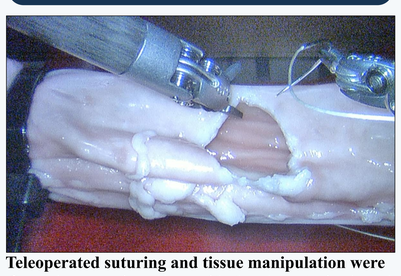
Ex-vivo colon

ZERO-SHOT COLON



Ground truth (black) vs. prediction (red)

EX VIVO ZERO-SHOT EVALUATION



Teleoperated suturing and tissue manipulation were performed on porcine colon, with the glove-trained model evaluated zero-shot without fine-tuning.

Quantitative Results:

Rubber-glove Test

RMSE (N)

Fx	Fy	Fz	Mean
1.15	0.54	1.22	0.97

RMSEs on the held-out test orientations.

Ex vivo Colon

Force-Change Direction Accuracy (%)

Fx	Fy	Fz	Mean
87.9	69.4	70.6	75.9

Balanced accuracy for force-change direction after 45-sample causal smoothing.

Discussion:

① Stereo endoscopic video supports 3D reconstruction and force estimation.

② Zero-shot transfer preserved force trends, but absolute force was inaccurate.

③ Tension-free colon limits force transmission to the sensor during suturing, motivating improved setups or alternative force-estimation methods.

④ Task- and tissue-specific adaptation and data are needed for reliable force-informed autonomy.

For more details, please see the full paper

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